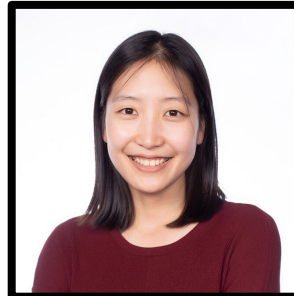


Language Models Cheat on Multiple-Choice Benchmarks. Good!

Nishant (that's me!)



Nishant: A bit about me!

- Ph.D. @ UMD, work on evals to make LLMs more helpful
- Eval work is often biased to be negative (AI can't do X, ...)
- Today, I'll be skeptical!
- If prior work says something is an “issue”, should we trust it?

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I'll be talking about multiple-choice evaluations, a popular NLP task

Multiple-Choice Question

Question: What is the capital of Italy?

(A) Rome

(B) Pisa

(C) Madrid

(D) None of the Above

Answer: (A)

Nishant: A bit about me!

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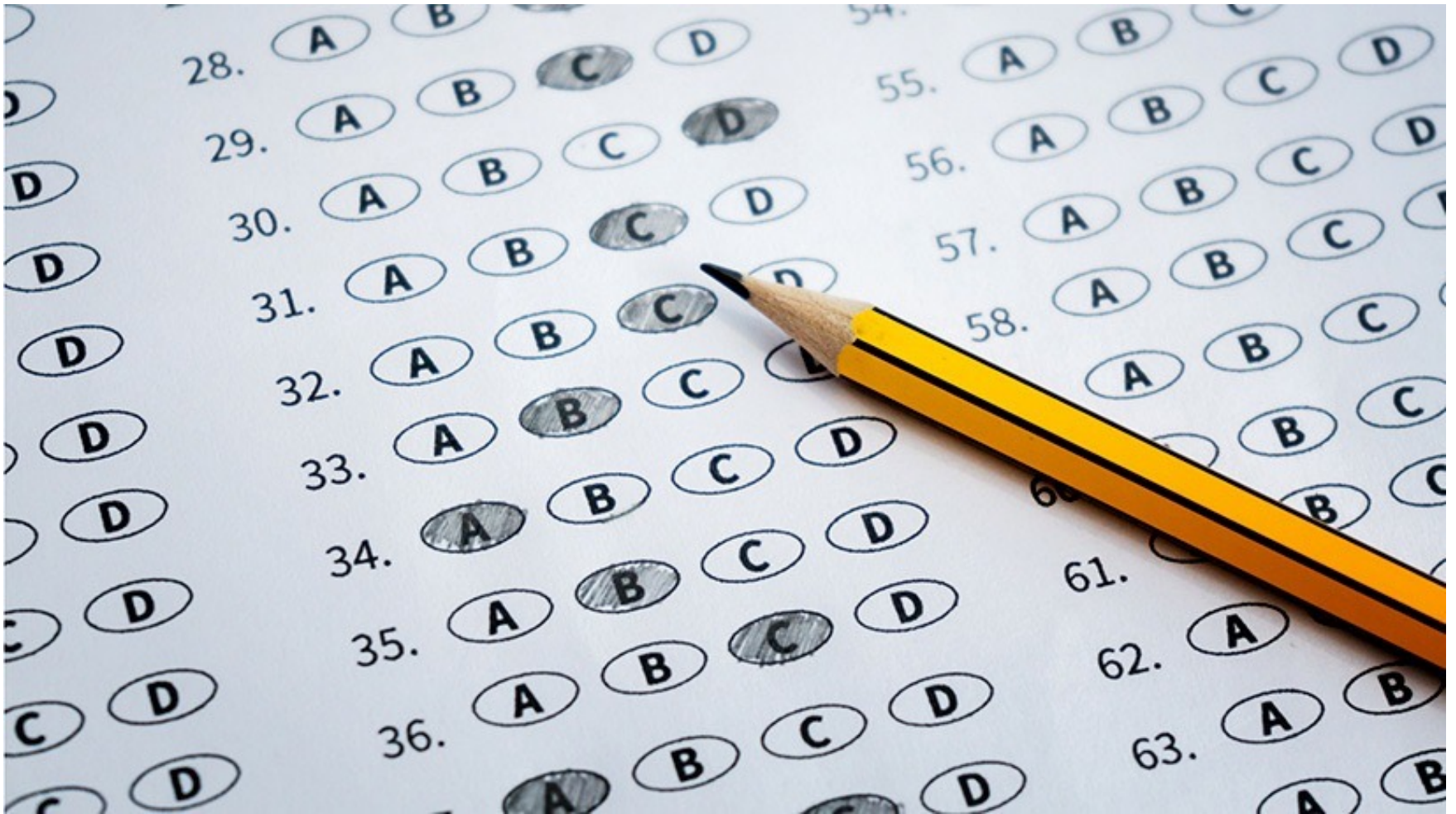
(C) Madrid

(D) None of the Above

Answer: (A)

And specifically, ways models can *cheat* on them

It's exam day!



But you “forgot” to study :(

This is your first question and it's hard...

Multiple-Choice Question

Jeremy was training for a charity walk and kept a steady pace of 3 kilometers per hour. If he started walking at 8:15 a.m. and took a break at 8:45 a.m., how far had he walked during that time?

- (A) 0.1km
- (B) 1.5km
- (C) 10km
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I have no clue...

Time to outsmart the teacher :)

Multiple-Choice Question

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1.5km is not too high or low, and it's a reasonable distance!

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1.5km is not too high or low, and it's a reasonable distance!

Papers find LLMs can do this on popular benchmarks!

Answer Matching Outperforms Multiple Choice

What Does My QA Model Know?

Improving Model Evaluation using SMART Filtering of Benchmark

Does Question Really Matter? The Attribution of Answer Bias in LLM Evaluation

**Artifacts or Abduction:
How Do LLMs Answer Multiple-Choice Questions Without the Question?**

Nishant Balepur¹ Abhilasha Ravichander² Rachel Rudinger¹
¹University of Maryland ²Allen Institute for Artificial Intelligence
{nbalepur, rudinger}@umd.edu, abhilashar@allenai.org

“Discriminative shortcuts can plague multiple-choice benchmarks”

But today, I hope to convince you that cheating in MCQA:

1. Isn't always a bad thing
2. Can help us build stronger evaluations

And we should change how we think about LLM capabilities

But today, I hope to convince you that cheating in MCQA:

1. Isn't always a bad thing

Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers

(ACL 2026)

Nishant Balepur^{1,2} Atrey Desai¹ Rachel Rudinger¹

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2. Can help us build stronger evaluations



BenchMarker: An Education-Inspired Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks

(ACL 2026)

**Nishant Balepur^{1,2} Bhavya Rajasekaran¹ Jane Oh¹ Michael Xie¹ Atrey Desai¹
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Studying model reasoning traces when cheating

RQ: Is solving multiple-choice questions without questions an issue?

- In older models that could do this (encoders), this RQ was hard to test
- But now: reasoning traces could give us insights into how models do this

We have 14 LLMs (GPT, Gemini, Claude) solve multiple-choice questions:

- With and without the question
- With and without chain-of-thought
- Across three MCQA benchmarks (ARC, MMLU, Super GPQA)

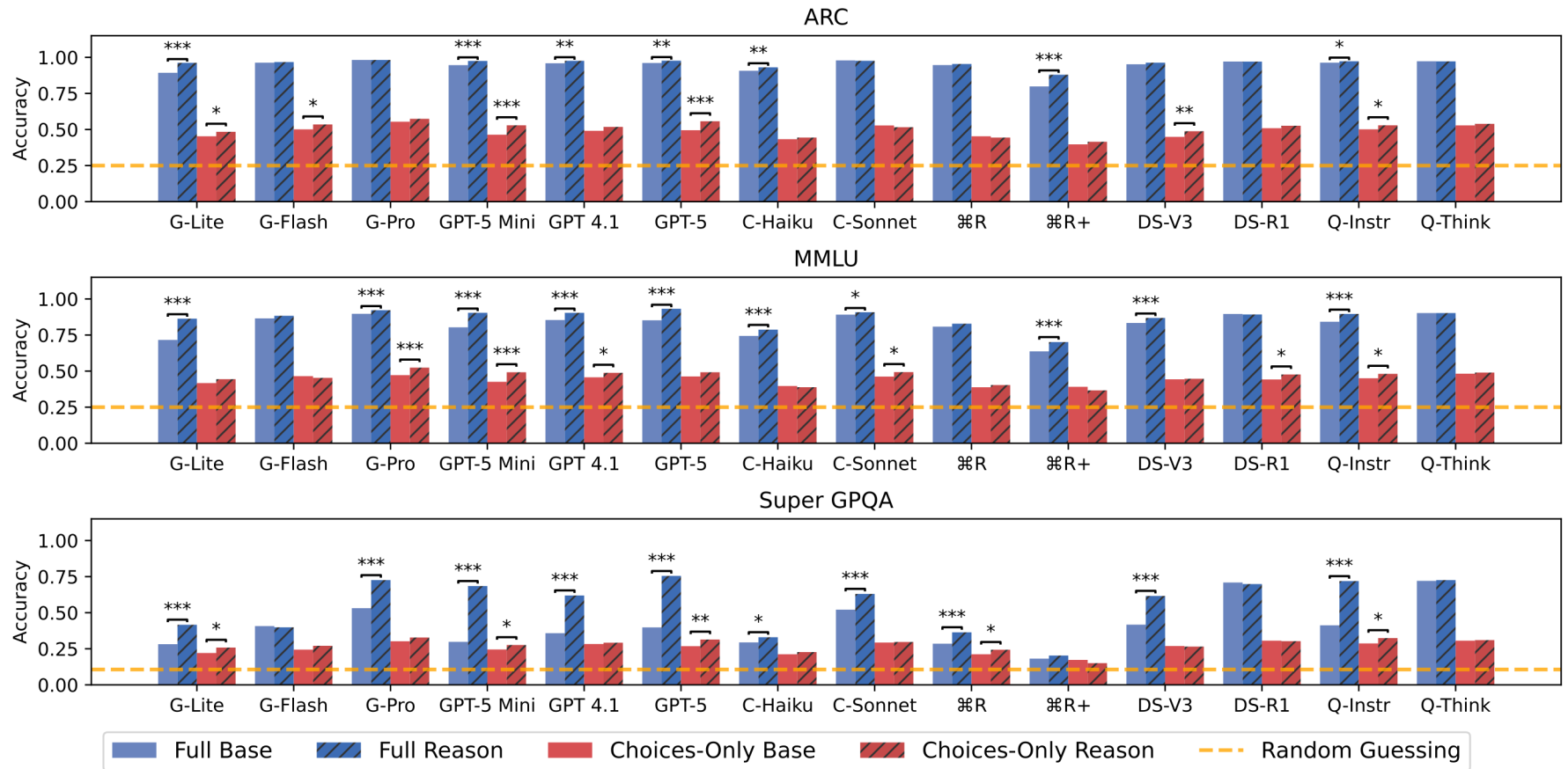
Multiple-Choice Question (no question)

~~Jeremy walked at an average rate of 3 km/h. How far had
Jeremy walked in 30 min?~~

- (A) 0.1 km (B) 1.5 km
(C) 10 km (D) 90 km

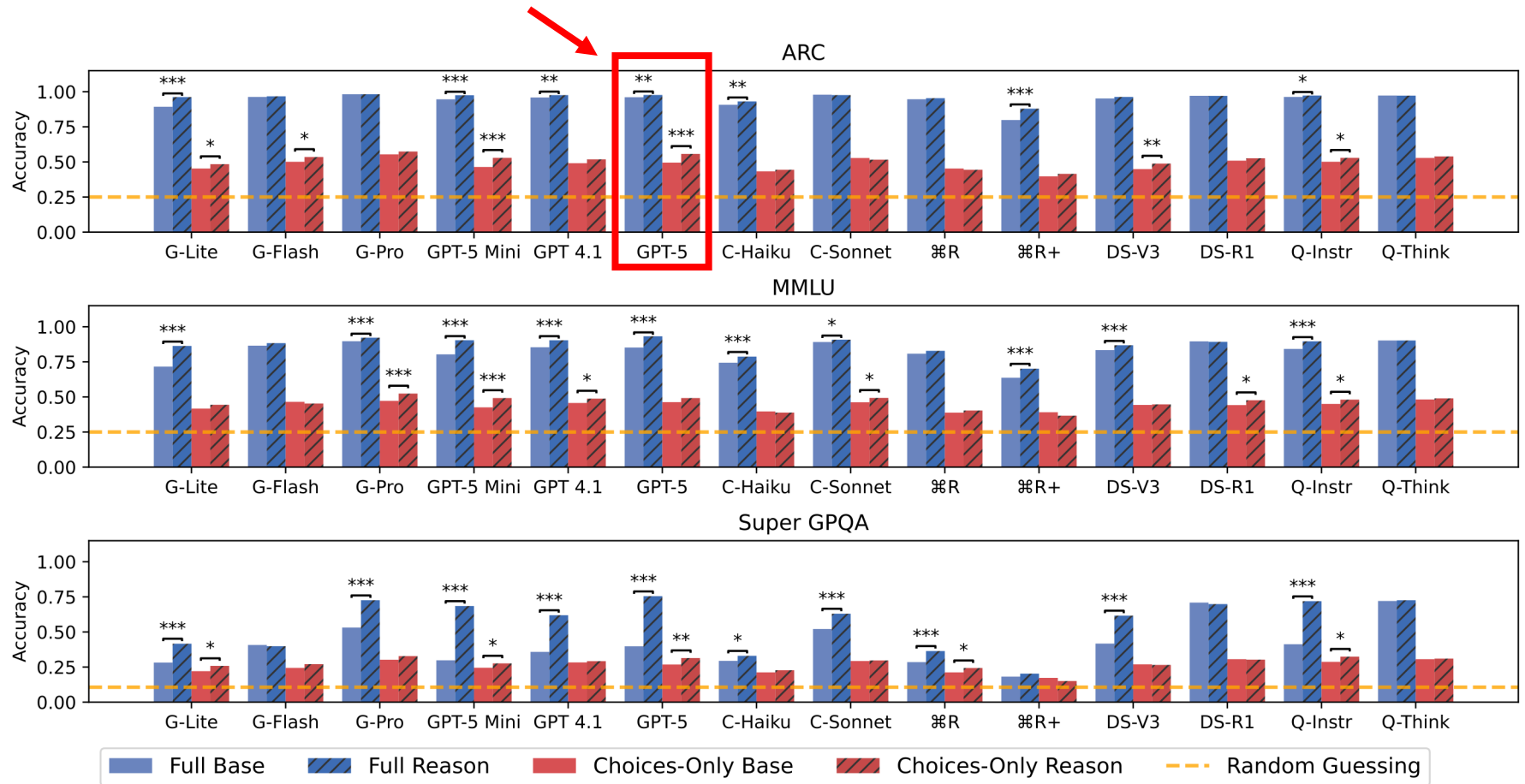
Reproduced: LLMs can answer MCQs without the Question!

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➤ LLMs largely exceed random guessing with the choices only

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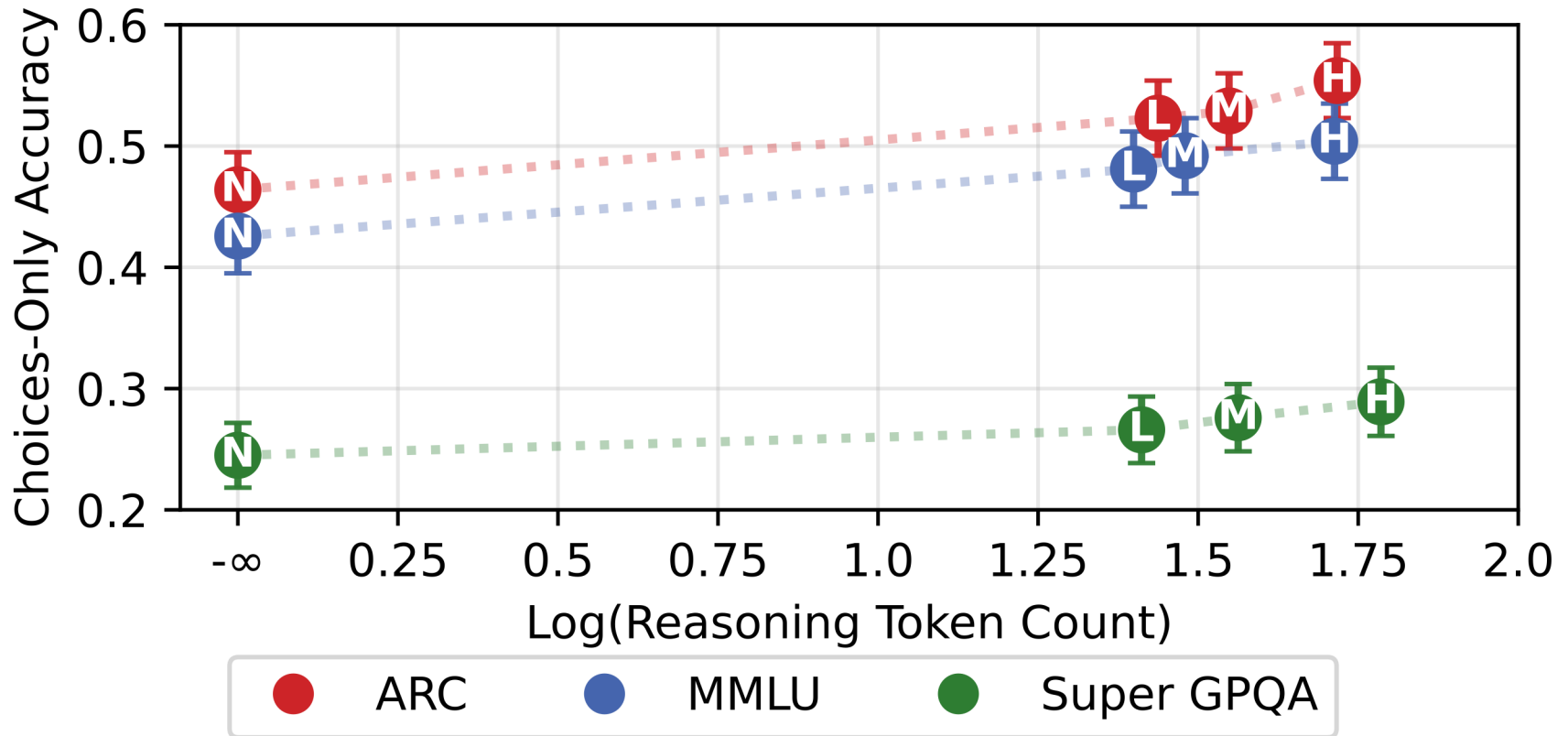


➤ LLMs largely exceed random guessing with the choices only

➤ GPT-5 reaches as high as 0.56 on ARC 🤖

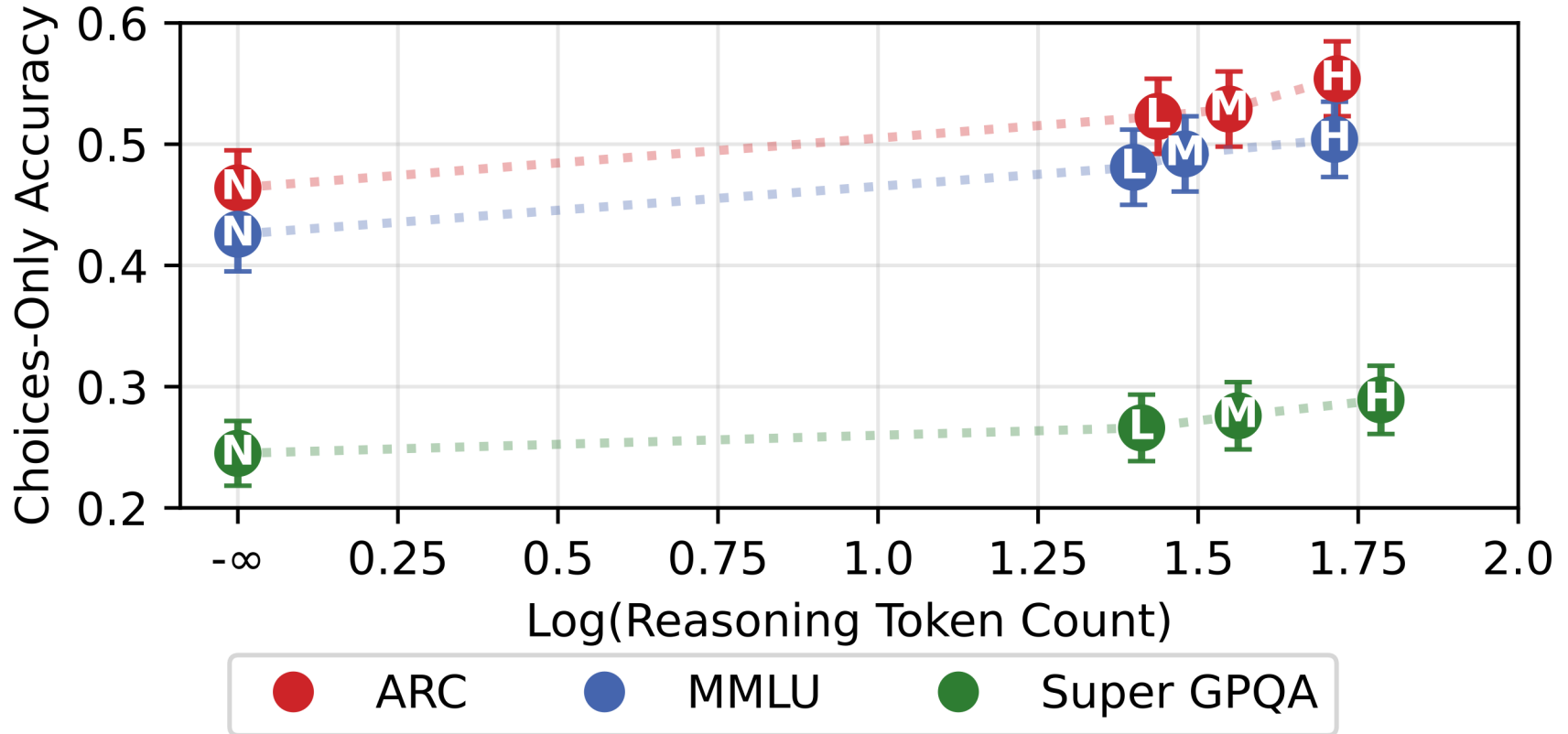
LLMs are not changing accuracy much after reasoning

LLMs are not changing accuracy much after reasoning



➤ Toggling **L**ow to **H**igh reasoning effort, there's not much of a change

LLMs are not changing accuracy much after reasoning



- Toggling **L**ow to **H**igh reasoning effort, there's not much of a change
- Perhaps a good sign that we can investigate LLM reasoning traces?

Analyzing reasoning traces shows interesting strategies!

Multiple-Choice Question

~~Jeremy walked at 3 km/h. How far had Jeremy walked in 30 min?~~

(A) 0.1 km (B) 1.5 km (C) 10 km (D) 90 km

Reasoning Trace

I'm given multiple-choice options but no question.

<think>

...The question likely concerns rate and time...

← Guess the original question

...1.5 is the messiest number...

← Use trivial shortcuts

Jogging for 15 min at 6 km/h is plausible...

← Rate choice plausibility

...0.1, 10, and 90 all could be related to powers of 10

← Find patterns in choices

</think>

Thus, my final answer is (B)

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</think>

Thus, my final answer is (B)

← Guess the original question

← Use trivial shortcuts

← Rate choice plausibility

← Find patterns in choices

➤ Some of these are arguably harder than the original MCQA task

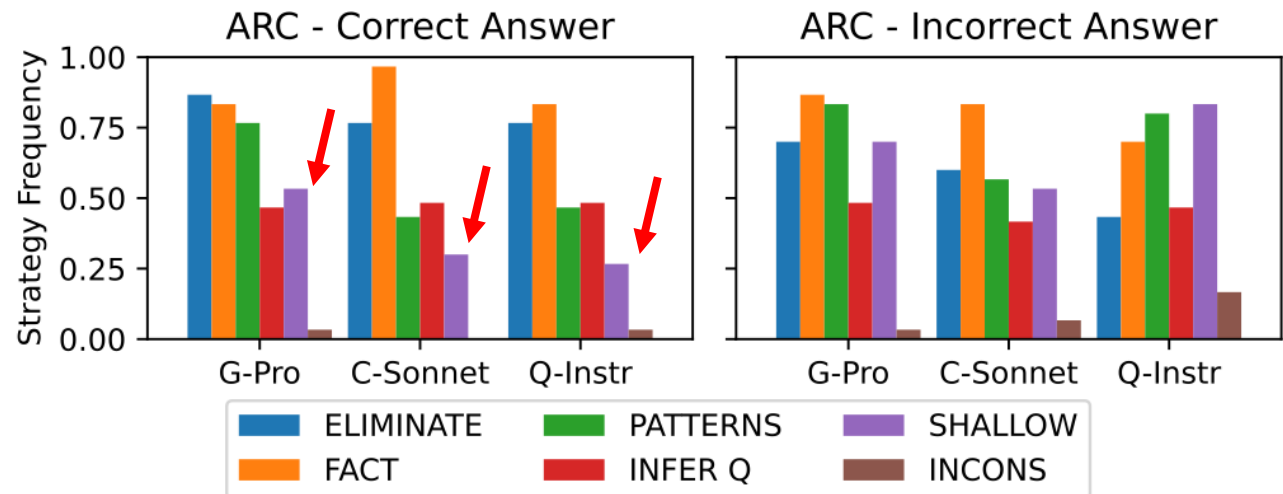


More formally, we develop a categorization of strategies

Two authors
review a subset of
traces, come up
with themes

Strategy	Example
ELIMINATE answers	<i>Spiders eat insects, not grass, and rabbits eat plants, not mice. This is a mess</i>
PATTERNS in choices	<i>Looking at these, I see patterns... three are non-renewable resources</i>
Recall a relevant FACT	<i>Scanning the options, (A) "need energy to survive" leaps out as a universal truth</i>
INFER the question	<i>I'm confident the question is asking about the best way to ensure reliable results</i>
SHALLOW shortcut	<i>1.5 is the only value with a "5". It's the "messiest"</i>

Models often use
non-shallow
strategies in
choices-only!



What does this mean for NLP evaluation?

Most work treats this choices-only success as a model/dataset issue

Answer Matching Outperforms Multiple Choice

What Does My QA Model Know?

Improving Model Evaluation using SMART Filtering of Benchmark

Does Question Really Matter? The Attribution of Answer Bias in LLM Evaluation

Boxi Cao, Ruotong Pan, Hongyu Lin, Xianpei Han*, Le Sun*

Chinese Information Processing Laboratory, Institute of Software
Chinese Academy of Sciences, Beijing, China
{caoboxi, hongyu, xianpei, sunle}@iscas.ac.cn

“Discriminative shortcuts can plague multiple-choice benchmarks”

But this isn't always true!

- Sometimes, models exploit dataset issues
- Sometimes, models use impressive strategies

This motivates: Can we disentangle these cases? 🙄

Pause for
questions... 🧑

But today, I hope to convince you that cheating in MCQA:

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2. Can help us build stronger evaluations



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Imagine you were taking this exam...

Multiple-Choice Exam

Question: What is the capital of Georgia?

Choices:

(A) Alabama

(B) Atlanta

(C) Madrid

(D) None of the Above

Answer:

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(B) Atlanta!

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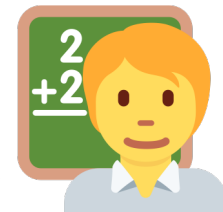
Answer: (B)



(B) Atlanta!



Wow, you must really know capitals!



Failure Case 1: Shortcuts!

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(A) isn't even a city and (C) isn't in the U.S., I guess (B)?

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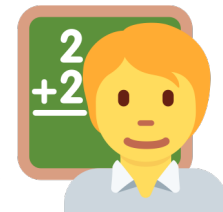
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Failure Case 2: Memorization!

Multiple-Choice Exam

Question: What is the capital of Georgia?

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Answer:



Failure Case 2: Memorization!

Multiple-Choice Exam

Question: What is the capital of Georgia?

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Answer: (B)



*I looked up last year's exam,
so (B)!*

Chegg®

Failure Case 2: Memorization!

Multiple-Choice Exam

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Choices:

(A) Alabama

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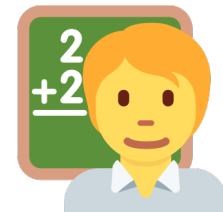


*I looked up last year's exam,
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Chegg[®]



Wow, you must really know capitals!



Failure Case 3: Writing Flaws!

Multiple-Choice Exam

Question: What is the capital of Georgia?

Choices:

(A) Alabama

(B) Atlanta

(C) Madrid

(D) None of the Above

Answer:



Failure Case 3: Writing Flaws!

Multiple-Choice Exam

Question: What is the capital of Georgia?

Choices:

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(B) Atlanta

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Answer: (B)



(D), Georgia's capital is Tbilisi!

Failure Case 3: Writing Flaws!

Multiple-Choice Exam

Question: What is the capital of Georgia?

Choices:

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(B) Atlanta

(C) Madrid

(D) None of the Above

Answer: (B)



(D), Georgia's capital is Tbilisi!

F
*see me
after class!*

You know nothing about capitals



In human exams, this doesn't happen too often

Multiple-Choice Exam

Question: What is the capital of Georgia?

Choices:

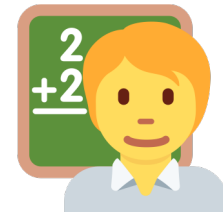
(A) Alabama

(B) Atlanta

(C) Madrid

(D) None of the Above

Answer: (B)



Reducing memorization

Multiple-Choice Exam

Question: What is the capital of Georgia?

Choices:

(A) Alabama

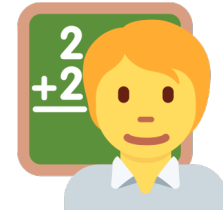
(B) ~~Atlanta~~ Madrid

(C) ~~Madrid~~ Atlanta

(D) None of the Above

Answer: ~~(B)~~ (C)

I'll make sure this isn't on Google



Eliminating obvious shortcuts

Multiple-Choice Exam

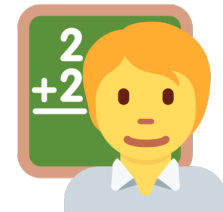
Question: What is the capital of Georgia?

Choices:

- (A) ~~Alabama~~ Macon
- (B) ~~Atlanta~~ ~~Madrid~~ Savannah
- (C) ~~Madrid~~ Atlanta
- (D) None of the Above

Answer: ~~(B)~~ (C)

Choices (A) and (B) are too guessable



Fixing poor writing

Multiple-Choice Exam

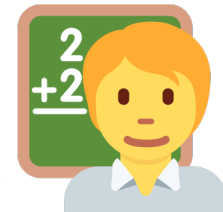
Question: What is the capital of Georgia, USA?

Choices:

- (A) ~~Alabama~~ Macon
- (B) ~~Atlanta~~ ~~Madrid~~ Savannah
- (C) ~~Madrid~~ Atlanta
- (D) None of the Above

Answer: ~~(B)~~ (C)

I'll clarify that I mean the state!



Unfortunately, NLP benchmarks don't follow this process

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Terrible Multiple-Choice Question 1

How many seconds would it most likely take to get to Mars?

(A) 40000000000000000000000000 seconds

(B) 40 seconds (C) 400 seconds (D) 4 seconds

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Terrible Multiple-Choice Question 2

Having a conversation in rooms with bare cement can be challenging because of:

- (A) dancing clown guys (B) bouncing sound waves
(C) problematic dingoes (D) egg cartons

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Having a conversation in rooms with bare cement can be challenging because of:

- (A) dancing clown guys (B) bouncing sound waves
(C) problematic dingoes (D) egg cartons

Confused NLP system



???

Let's make MCQA benchmarks rigorous!

MCQA benchmarks track NLP progress, but lack quality control

- We surveyed 32 datasets: 28% did no quality control
- Despite valuing MCQA's similarity to student test taking, we aren't scrutinizing them for flaws like educators do!
- The first step towards correction is detection, but this is tedious...

RQ: Can we try to automate this process?



BenchMarker: Detecting Multiple-Choice Question Flaws

Extending ideas from paper #1, we use LLM judges to detect three issues:

BenchMarker: Detecting Multiple-Choice Question Flaws

Extending ideas from paper #1, we use LLM judges to detect three issues:

Which of These Best Describes Multiple Choice Evaluation with LLMs?
A) Forced B) Flawed C) Fixable D) All of the Above

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Rachel Rudinger

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University of Maryland

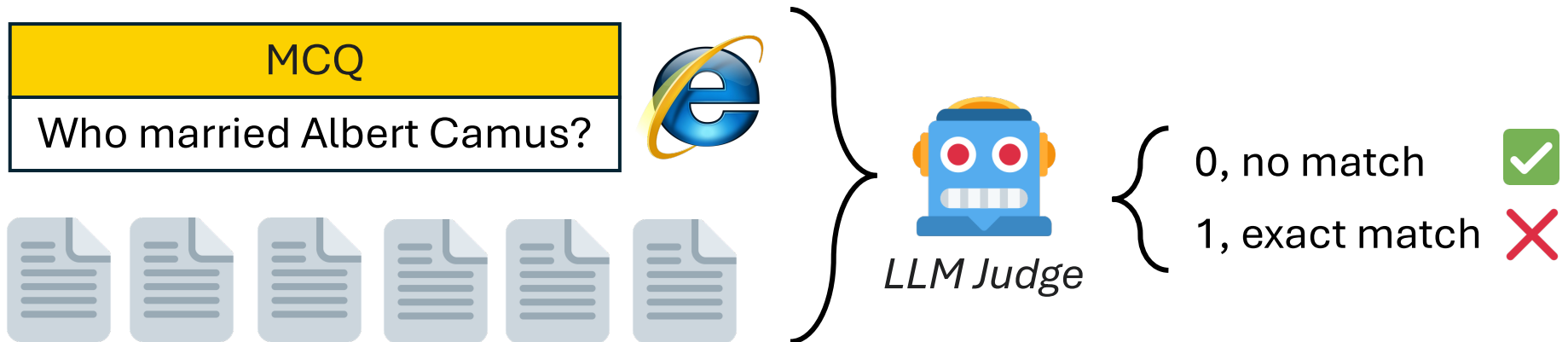
jbg@umiacs.umd.edu

Has way more ideas from education research on improving MCQA
benchmarks! (at ACL 2025)

BenchMarker: Detecting Multiple-Choice Question Flaws

Extending ideas from paper #1, we use LLM judges to detect three issues:

- **Contamination:** Items that exist *exactly* on the internet [1]



(Internet is our proxy for pre-training data)

BenchMarker: Detecting Multiple-Choice Question Flaws

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- **Contamination:** Items that exist *exactly* on the internet
- **Shortcuts:** LLMs can cheat on multiple-choice questions

MCQ

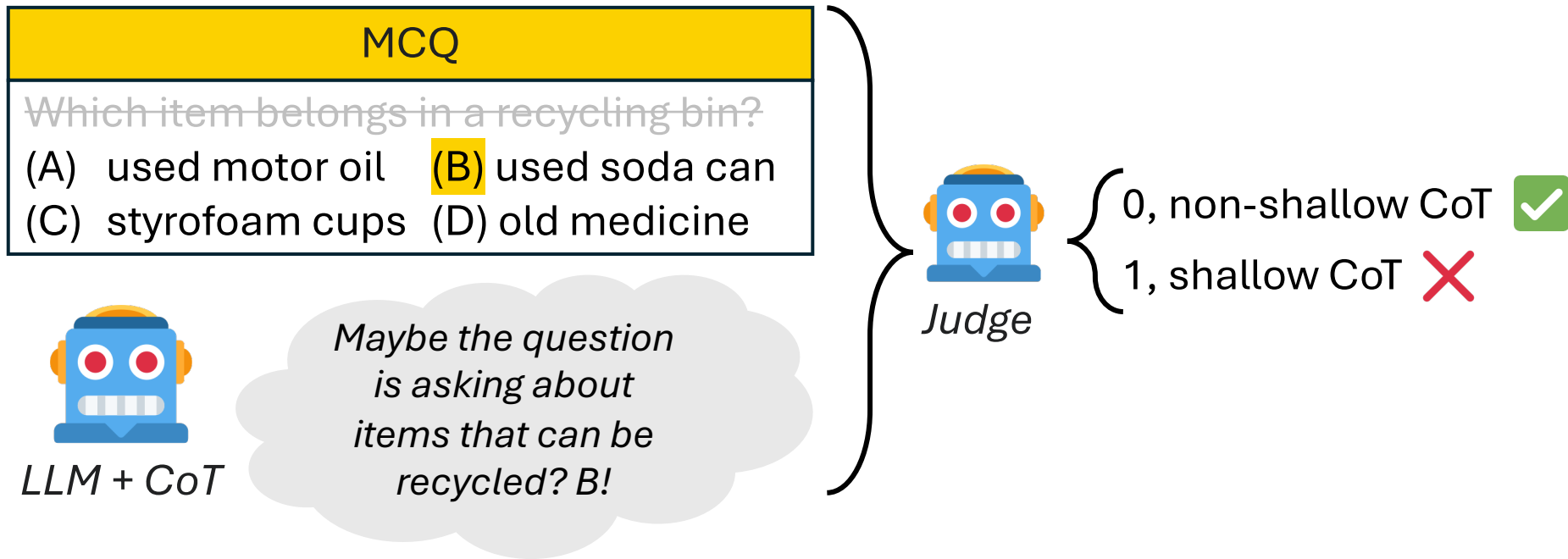
Which item belongs in a recycling bin?

- (A) used motor oil (B) used soda can
(C) styrofoam cups (D) old medicine

BenchMarker: Detecting Multiple-Choice Question Flaws

Extending ideas from paper #1, we use LLM judges to detect three issues:

- **Contamination:** Items that exist *exactly* on the internet
- **Shortcuts:** LLMs can cheat on multiple-choice questions [1, 2]



[1] [How Do LLMs Answer Multiple-Choice Questions Without the Question?](#) (ACL 24)

[2] [Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers](#) (ACL 26)

BenchMarker: Detecting Multiple-Choice Question Flaws

Extending ideas from paper #1, we use LLM judges to detect three issues:

- **Contamination:** Items that exist *exactly* on the internet
- **Shortcuts:** LLMs can cheat on multiple-choice questions
- **Writing Errors:** Flaws from a 19-rule rubric from education [1, 2]

Rubric of Rules from Education
1. Do not include “All of the above”
2. Avoid ambiguous terms like “mostly”
...
19. Keep grammar consistent in choices

MCQ
...



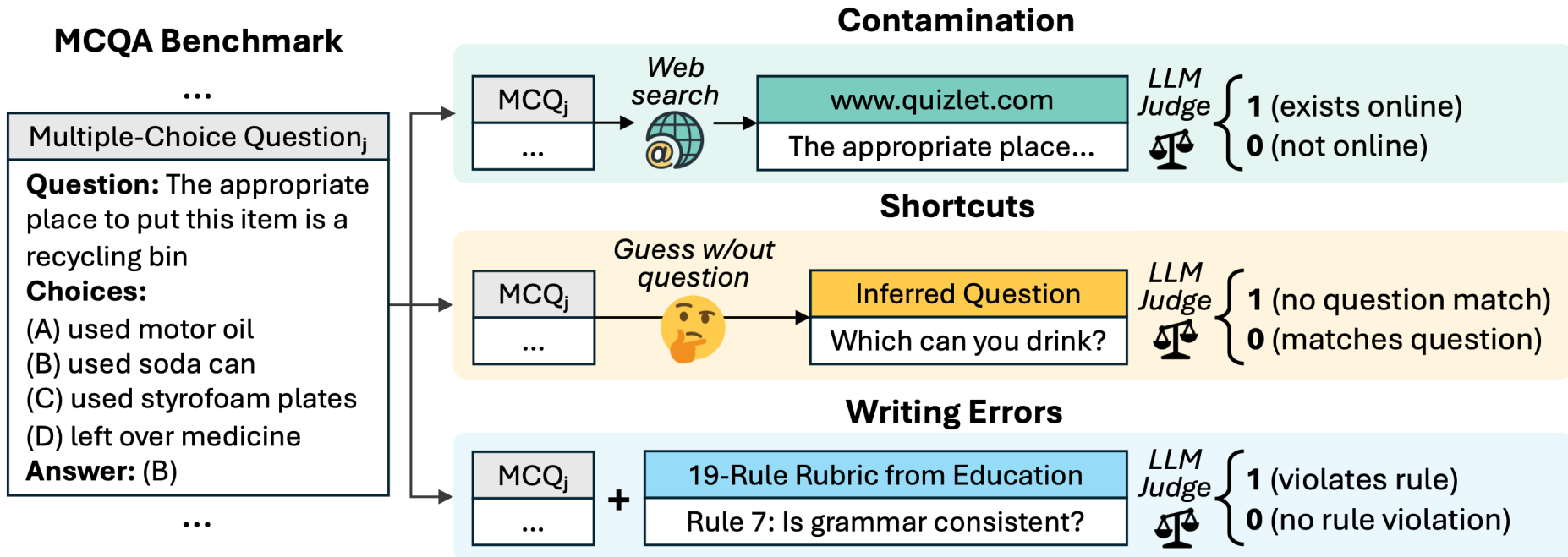
LLM Judge

{ 0, passed 
1, violated 

[1] [Impact of item-writing flaws in multiple-choice questions on student achievement](#)

[2] [Plausibly Problematic Questions in Multiple-Choice Benchmarks](#) (EMNLP 25 findings)

BenchMarker: Detecting Multiple-Choice Question Flaws



Report Card
Contamination: 0.21 > Item <i>j</i> was found online
Shortcuts: 0.32 > Item <i>j</i> has a shortcut: the right answer is the only drink
Writing Errors: 0.45 > Item <i>j</i> R7: "This item" has a grammar mismatch - "plates"

- Overall scores and per-item breakdown
- Rationales for model predictions
- Implemented in InspectAI (UI, logging, ...)

Validating BenchMarker: Human Annotations

We collect a validation set of 8042 labels to pick LLM judges for BenchMarker

- 3,419 labels are from us (the authors!)
 - Author A labels MCQs by doing the same task as LLM judges
 - For contamination, they can access search engines
 - Author B labels a subset of 950 → over 80% agreement per metric
- 4,123 writing error labels from [Costello et al. \(2018a\)](#) → out-of-domain data

Validating BenchMarker: Which Search Engine to Use?

Validating BenchMarker: Which Search Engine to Use?

Method	Accuracy	F1 Score	Cohen's κ
Google + Gemini Pro	0.70*	0.65	0.41
Google + GPT-5	0.71*	0.68	0.44
Google + Claude Sonnet	0.69*	0.64	0.40
Brave + GPT-5	0.54	0.28	0.14
Perplexity + GPT-5	0.64	0.53	0.31
Exa + GPT-5	0.59	0.43	0.22
Tavily + GPT-5	0.58	0.41	0.20
Serper + GPT-5	0.68	0.65	0.36
Random (50/50)	0.50	0.52	0.00
Always Not Flawed	0.46	0.00	0.00
Always Flawed	0.54	0.70	0.00

Validating BenchMarker: Which Search Engine to Use?

Method	Accuracy	F1 Score	Cohen's κ
Google + Gemini Pro	0.70*	0.65	0.41
Google + GPT-5	0.71*	0.68	0.44
Google + Claude Sonnet	0.69*	0.64	0.40
Brave + GPT-5	0.54	0.28	0.14
Perplexity + GPT-5	0.64	0.53	0.31
Exa + GPT-5	0.59	0.43	0.22
Tavily + GPT-5	0.58	0.41	0.20
Serper + GPT-5	0.68	0.65	0.36
Random (50/50)	0.50	0.52	0.00
Always Not Flawed	0.46	0.00	0.00
Always Flawed	0.54	0.70	0.00

➤ GPT-5 is the most reliable backbone LLM

Validating BenchMarker: Which Search Engine to Use?

Method	Accuracy	F1 Score	Cohen's κ
Google + Gemini Pro	0.70*	0.65	0.41
Google + GPT-5	0.71*	0.68	0.44
Google + Claude Sonnet	0.69*	0.64	0.40
Brave + GPT-5	0.54	0.28	0.14
Perplexity + GPT-5	0.64	0.53	0.31
Exa + GPT-5	0.59	0.43	0.22
Tavily + GPT-5	0.58	0.41	0.20
Serper + GPT-5	0.68	0.65	0.36
Random (50/50)	0.50	0.52	0.00
Always Not Flawed	0.46	0.00	0.00
Always Flawed	0.54	0.70	0.00

- GPT-5 is the most reliable backbone LLM
- Google is the most reliable search engine

Validating BenchMarker: Which LLM Judges to Use?

Validating BenchMarker: Which LLM Judges to Use?

Method	<i>Shortcuts</i>			<i>Writing (In Domain, NLP)</i>		
	Accuracy	F1 Score	Cohen's κ	Accuracy	F1 Score	Cohen's κ
Gemini 2.5 Lite	0.76	0.74	0.51	0.68	0.56	0.35
Gemini 2.5 Flash	0.69	0.68	0.41	0.79*	0.62	0.47
Gemini 2.5 Pro	0.70	0.69	0.43	0.82*	0.66	0.53
GPT-5 Nano	0.68	0.65	0.38	0.74	0.39	0.22
GPT-5 Mini	0.77	0.72	0.53	0.75	0.55	0.38
GPT-5	0.82*	0.75	0.61	0.81 *	0.63	0.50
Claude 4.5 Haiku	0.78*	0.73	0.55	0.72	0.58	0.38
Claude 4.5 Sonnet	0.81*	0.75	0.59	0.79*	0.63	0.48
Command R	0.74	0.72	0.50	0.77	0.53	0.38
Command R+	0.72	0.70	0.46	0.76	0.53	0.36
Qwen-3 0.6B	0.71	0.64	0.39	0.73	0.05	0.00
Qwen-3 1.7B	0.62	0.64	0.31	0.76	0.25	0.16
Qwen-3 4B	0.42	0.55	0.05	0.73	0.49	0.32
Qwen-3 8B	0.57	0.61	0.24	0.70	0.54	0.33
Qwen-3 14B	0.61	0.64	0.30	0.71	0.55	0.34
Qwen-3 32B	0.73	0.71	0.48	0.71	0.55	0.35
Gemma-3 4B	0.50	0.59	0.16	0.63	0.46	0.20
Gemma-3 12B	0.61	0.64	0.31	0.75	0.04	0.03
Gemma-3 27B	0.74	0.72	0.50	0.75	0.03	0.02
LLaMA-3.2 1B	0.56	0.30	0.00	0.47	0.38	0.03
LLaMA-3.2 3B	0.58	0.62	0.26	0.68	0.37	0.16
LLaMA-3.1 8B	0.49	0.58	0.14	0.71	0.38	0.19
LLaMA-3.1 70B	0.61	0.62	0.28	0.64	0.47	0.22

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➤ Open-weight LLMs lag behind (we tried ensembling, calibration, ...)

Applying BenchMarker to benchmark benchmarks

Now that we have an auditing toolkit, let's apply it to benchmarks!

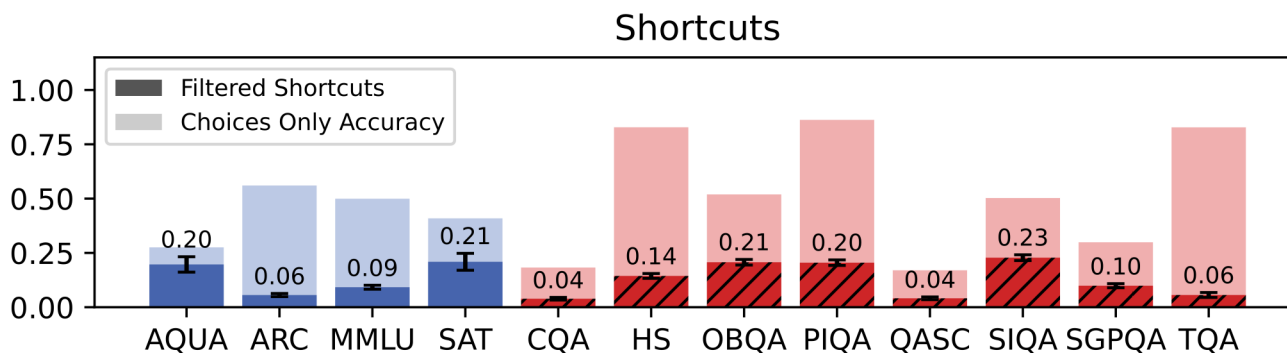
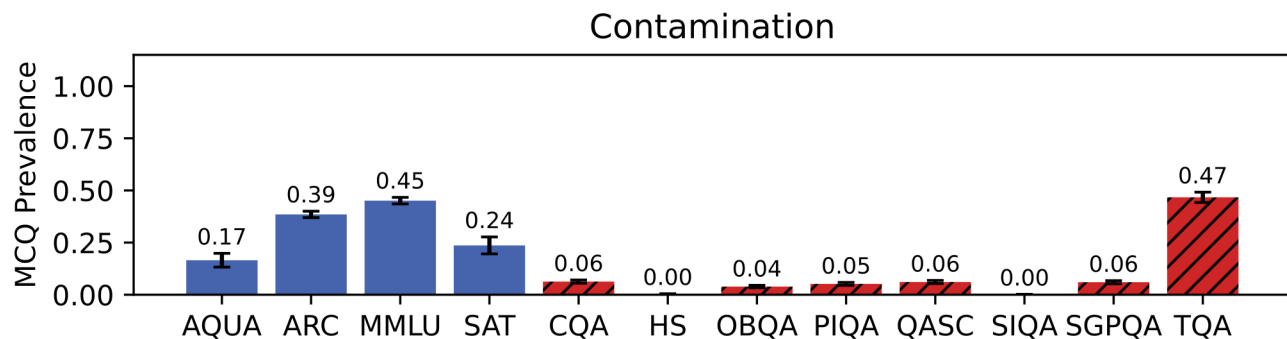
Dataset	Domain	Difficulty	Creation Strategy
Algebra Question Answering (Zhong et al., 2024, AQUA)	Math	Graduate	Student Exams
AI2 Reasoning Challenge (Clark et al., 2018, ARC)	Science	Elementary	Student Exams
CommonsenseQA (Talmor et al., 2019, CQA)	Commonsense	General	Crowdworkers
HellaSwag (Zellers et al., 2019, HS)	Commonsense	General	Model Generated
Multitask Language Understanding (Hendrycks et al., 2021, MMLU)	Multi-Subject	College	Student Exams
Open Book Question Answering (Mihaylov et al., 2018, OBQA)	Science	Elementary	Crowdworkers
Physical Interaction Question Answering (Bisk et al., 2020, PIQA)	Commonsense	General	Crowdworkers
Question Answering via Sentence Composition (Khot et al., 2020, QASC)	Science	Elementary	Crowdworkers
Scholastic Aptitude Test (Zhong et al., 2024, SAT)	Math	High School	Student Exams
Social Intelligence Question Answering (Sap et al., 2019, SIQA)	Commonsense	General	Crowdworkers
Super Google Proof Question Answering (Du et al., 2025, SGPQA)	Multi-Subject	Graduate	Expert+Model
Truthful Question Answering (Lin et al., 2021, TQA)	Commonsense	General	Author-Written

Table 1: The MCQA benchmarks we mainly explore. We audit MCQs across varied domains, difficulties, and creation strategies.

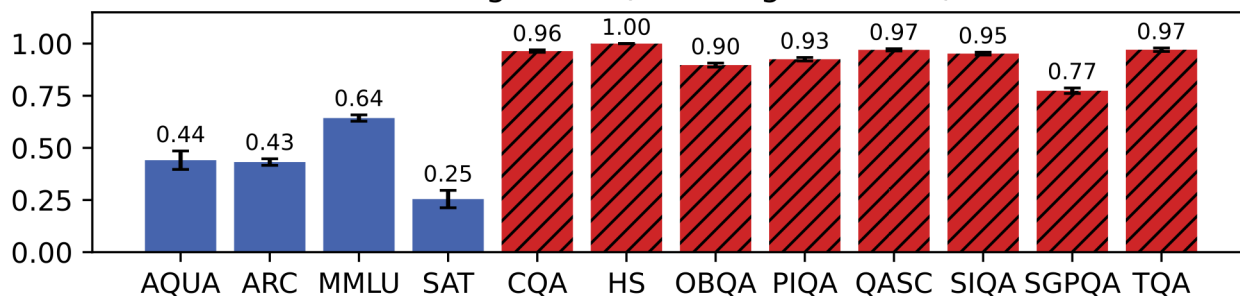
We select popular MCQA benchmarks across varied:

- Domains (math, science, commonsense, ...)
- Difficulty (elementary, high school, graduate, ...)
- Creation strategies (student exams, crowdworkers, model-generated, ...)

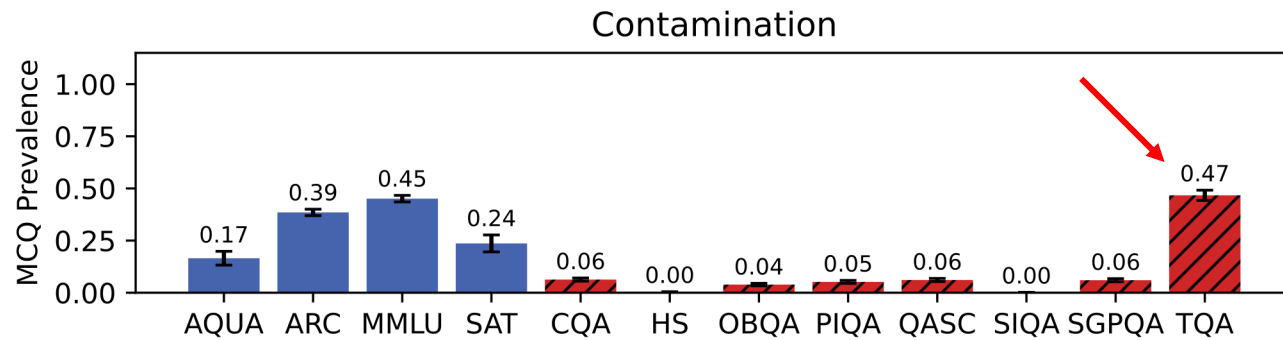
BenchMarker can detect flaws in NLP benchmarks



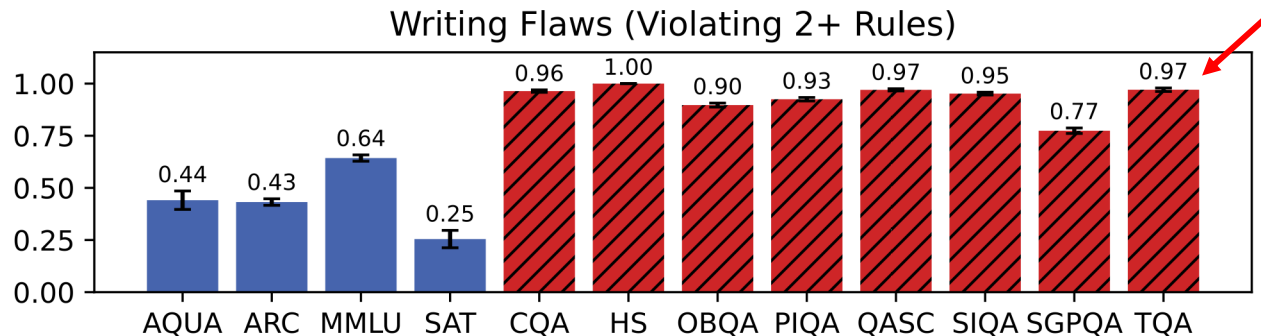
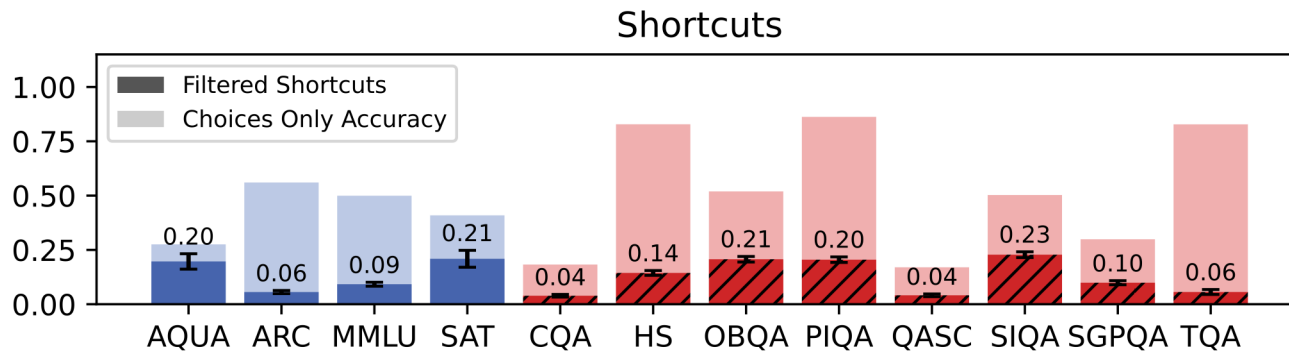
Writing Flaws (Violating 2+ Rules)



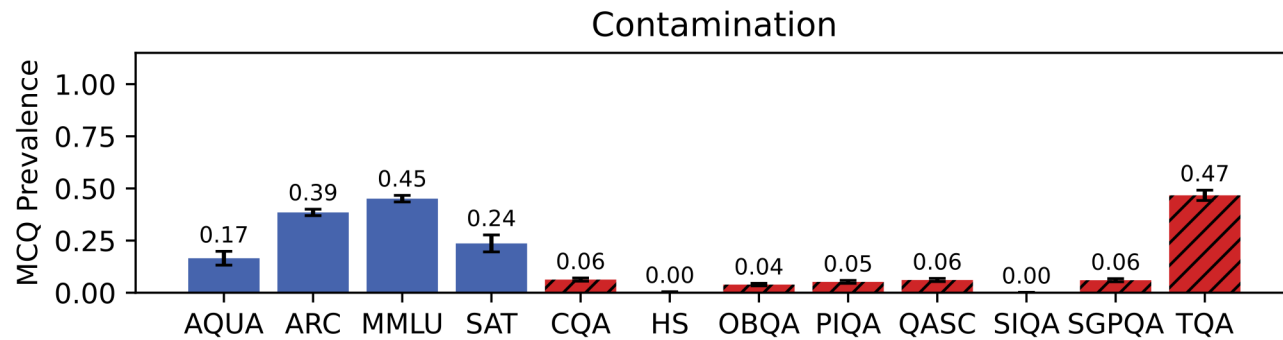
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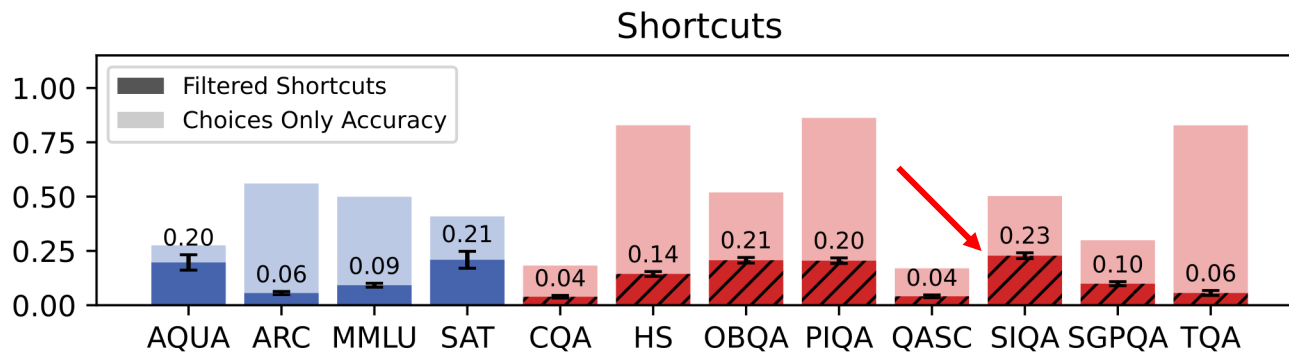
➤ Many NLP benchmarks are flawed!



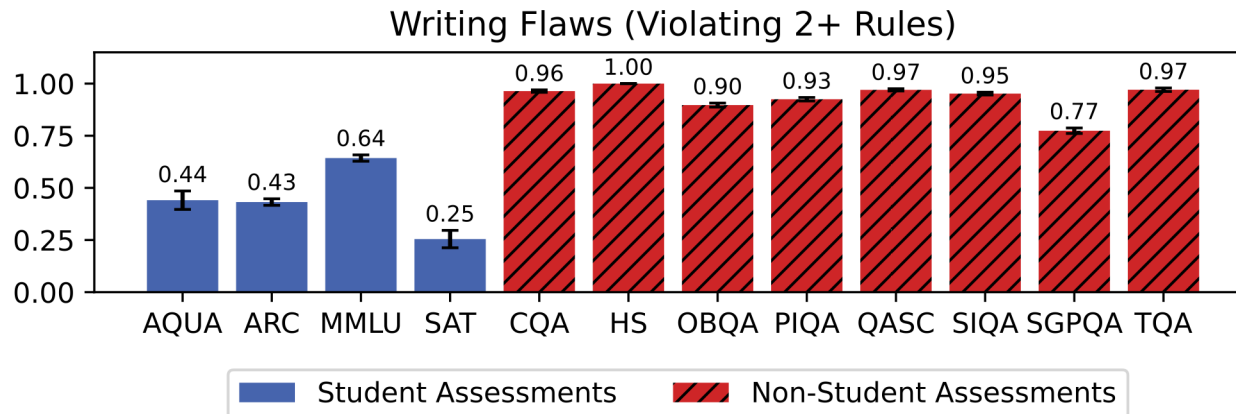
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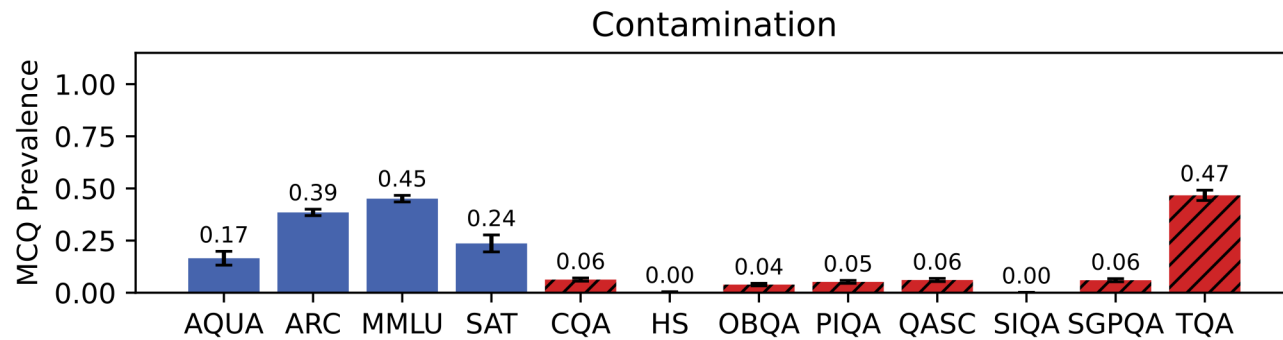
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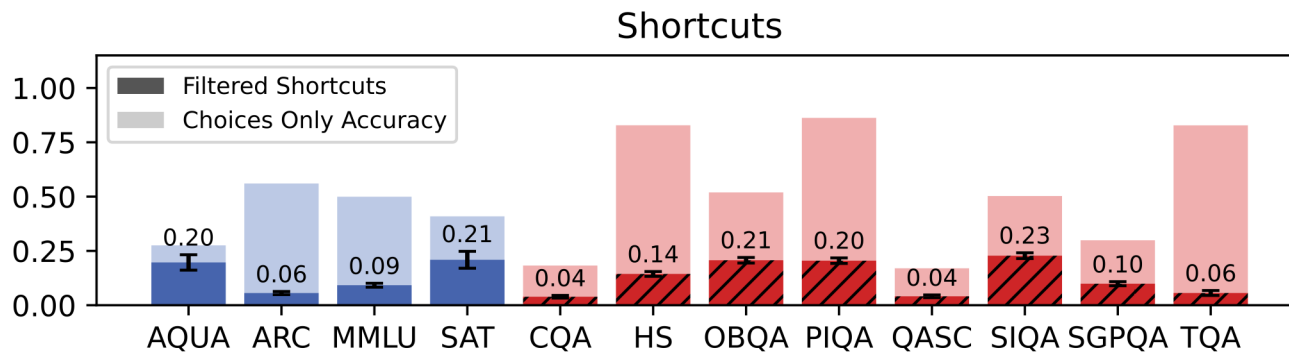
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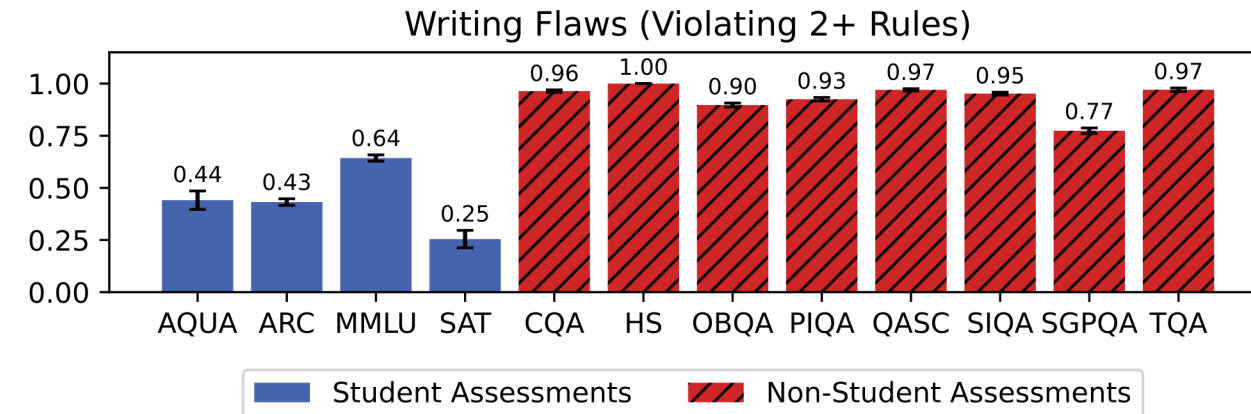
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➤ Many NLP benchmarks are flawed!



➤ Problematic shortcuts aren't that common



➤ Benchmarks from real exams are higher-quality than NLP-generated exams

And these flaws impact NLP evaluation!

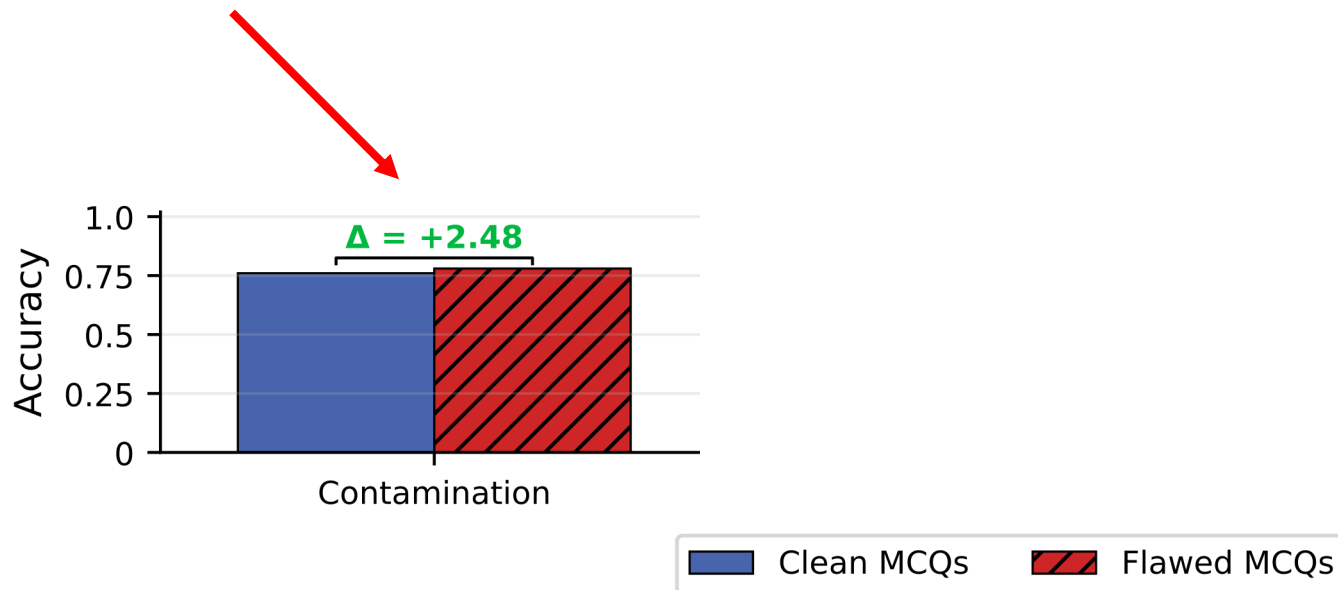
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We evaluate 10 test-taker LLMs on the clean + flawed MCQs:

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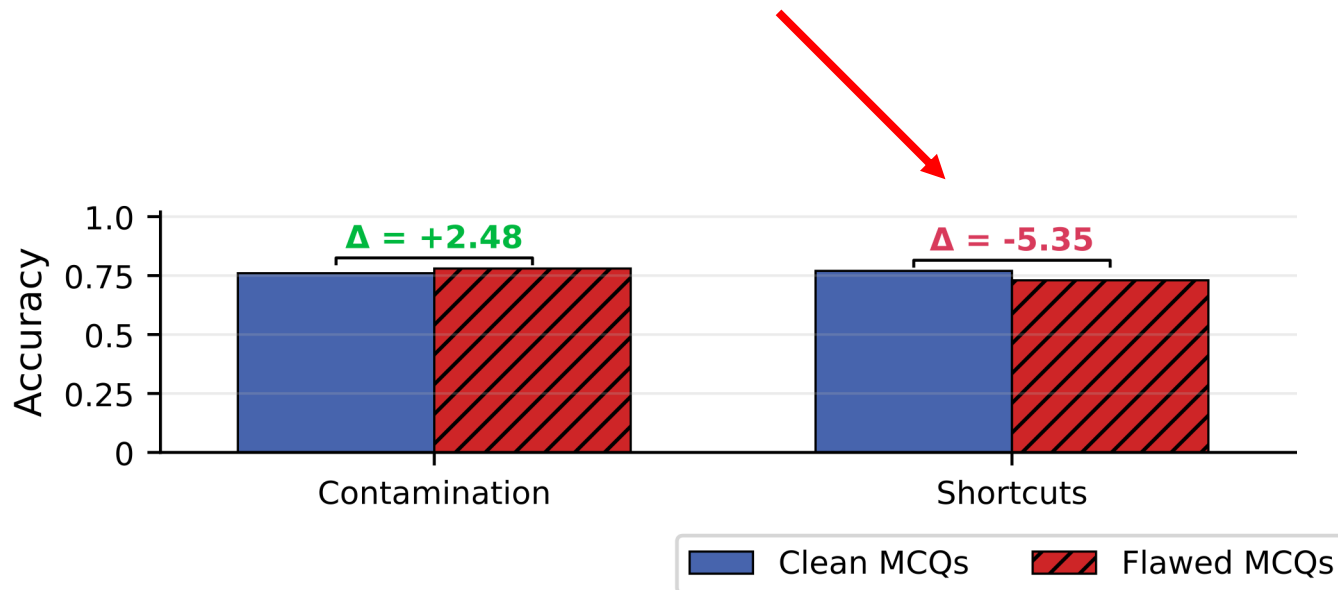
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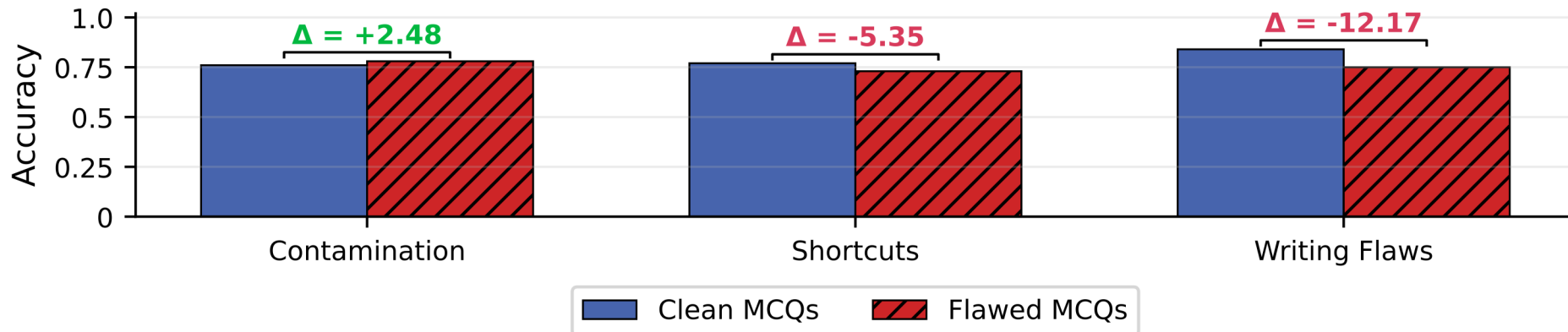
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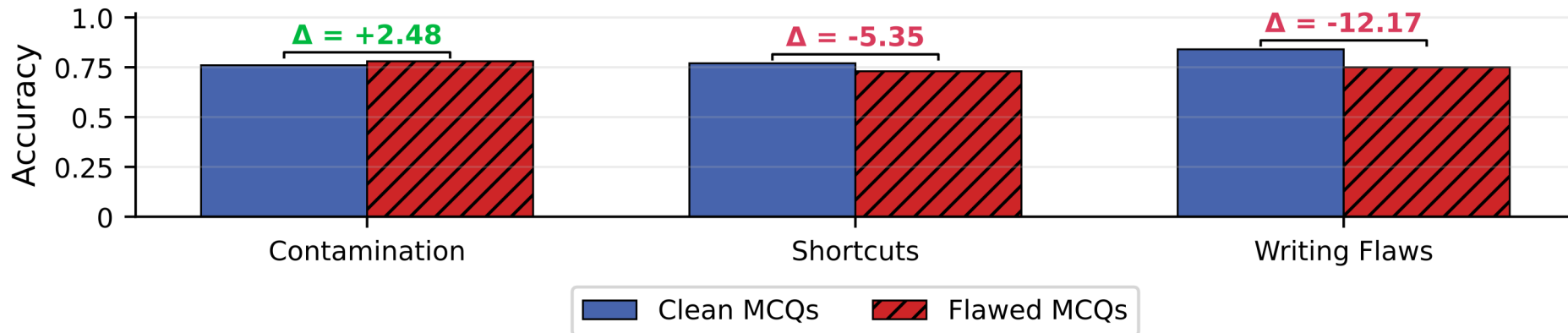
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Are we under-estimating models?

(I didn't show, but this also slightly impacts model ranks)

Is it simple for LLMs to fix these issues?

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We prompt LLMs to try and fix one dataset's writing flaws (TruthfulQA)

Model

GPT-5.2

Claude Sonnet

Qwen 235B

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➤ LLM's can't reduce all errors

Model	IP(2+ Errors)
GPT-5.2	0.97 → 0.64
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Simple prompting alone is not enough!

What about prior rewrites? MMLU-Pro?

MMLU-Pro rewrote MMLU with LLMs to make MCQs “harder”

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MMLU-Pro rewrote MMLU with LLMs to make MCQs “harder”

➤ Accuracy does drop!

Dataset	Contam.	Shortcuts	Writing	Accuracy
MMLU	0.45	0.12	0.12	0.79
MMLU Pro	0.24	0.12	0.15	0.63

What about prior rewrites? MMLU-Pro?

MMLU-Pro rewrote MMLU with LLMs to make MCQs “harder”

- Accuracy does drop!
- But writing flaws increase 🤔

Dataset	Contam.	Shortcuts	Writing	Accuracy
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How can we progress?

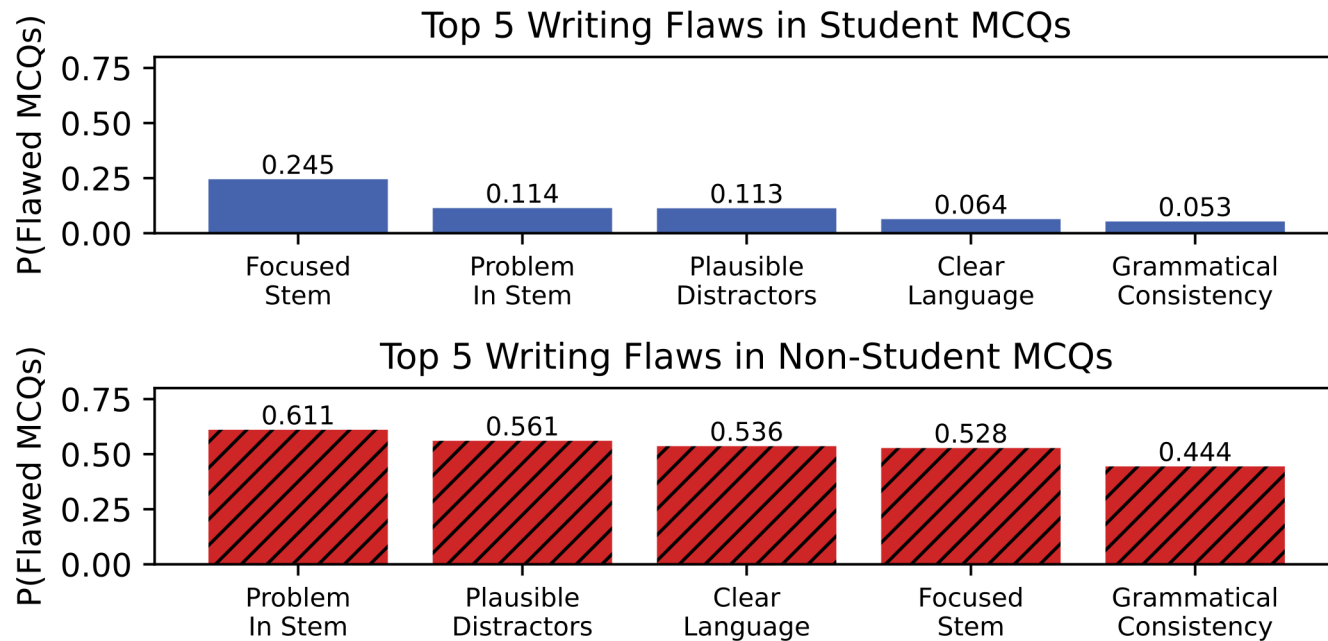
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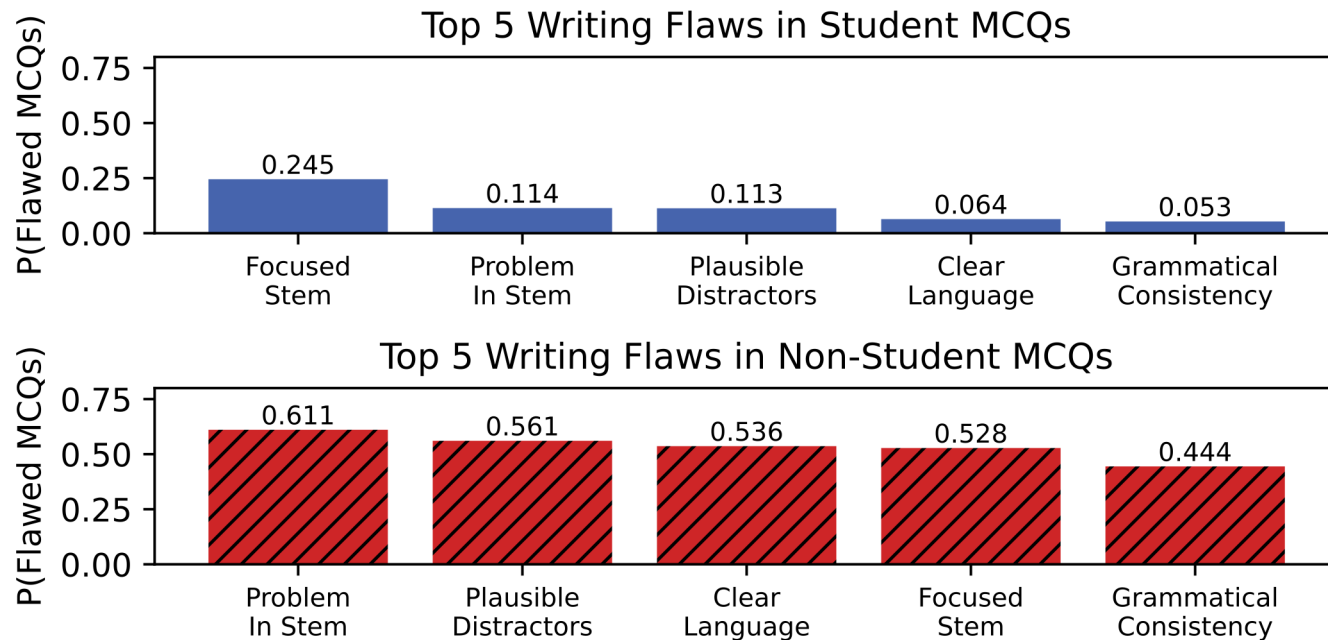
- Benchmarks derived from real student exams + NLP-generated exams have similar flaw distributions!



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NLP + education researchers can work together!

What did we learn? 🤔

1. Cheating isn't always a bad thing

Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers

(ACL 2026)

Nishant Balepur^{1,2} Atrey Desai¹ Rachel Rudinger¹

¹University of Maryland ²New York University

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2. Cheating help us build stronger evals

BenchMarker: An Education-Inspired Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks

(ACL 2026)

**Nishant Balepur^{1,2} Bhavya Rajasekaran¹ Jane Oh¹ Michael Xie¹ Atrey Desai¹
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LLMs can have abilities stronger than our evals 💪

Reasoning traces as a lens to find abilities? 🔍

2. Cheating help us build stronger evals

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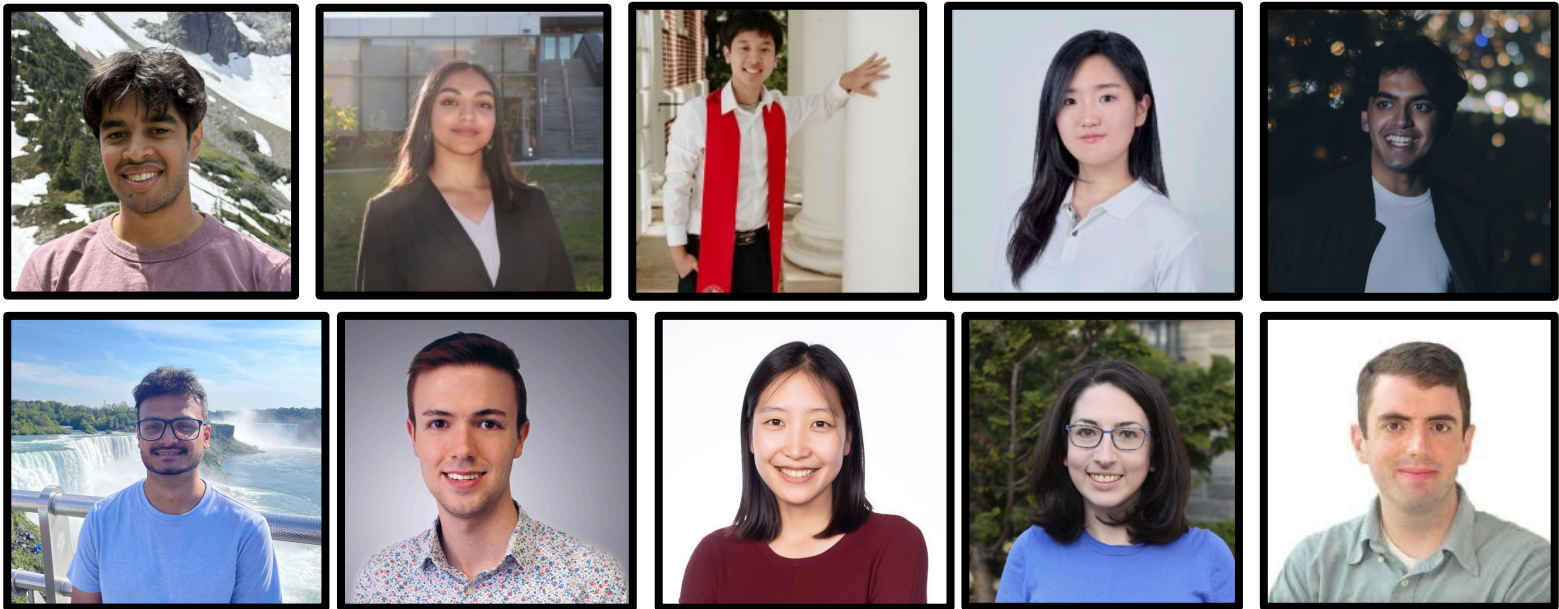
nbalepur@umd.edu jbg@umiacs.umd.edu

Benchmarks suck, but auditing is tractable 🕵️

Adversarial evaluations can help us find issues 😈

Thank You :)

Nishant (that's me!)



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 X.com: [@NishantBalepur](https://twitter.com/NishantBalepur)

 Website: <https://www.nbalepur.github.io>